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Ulva prolifera (Chlorophyta): a suitable material to remove Cd^{2+} from aquatic environments

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aquatic environment; cadmium; photosynthesis; *Ulva prolifera*.

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Abstract

Heavy metal pollution in aquatic environments is a significant environmental problem. Marine macroalgae are novel materials for the removal of heavy metals from water, aided by their large biomass and the feature of easy collection. In this study, *Ulva prolifera* O.F.Müller was used to remove cadmium ions (Cd^{2+}) from the aquatic environment. We examined the Cd^{2+} absorption capacity of the thalli in a factorial experiment using fresh thalli cultivated in seawater, freshwater, water containing domestic garbage, water containing industrial waste and elutriant of soil with additional Cd^{2+}. In addition, changes in photosynthetic activity of thalli were investigated. A significant decrease of maximum quantum yield and effective photochemical quantum yield of PSII (YII) indicated Cd^{2+} was transported into the cells and seriously inhibited photosynthesis. Furthermore, the Cd^{2+} absorption capacity was relatively high. The results indicated the suitability of *U. prolifera* for removing Cd^{2+} from aquatic environments of different backgrounds.

Introduction

Heavy metal pollution is becoming a worldwide, intractable problem. The main sources of heavy metal pollution are a consequence of human behaviour, including mining, industrial emission, waste disposal and sewage sludge. The heavy metals cadmium (Cd), lead (Pb), copper (Cu), zinc (Zn) and nickel (Ni) are among the most common pollutants in industrial effluent. They are toxic to humans, animals and plants, with severe negative effects on the growth and development of organisms (Athanasiadis & Helmreich 2005). Heavy metals are often enriched in organisms via the food chain, rather than being degraded. In particular, heavy metal pollution in the aquatic environment is a great threat to human health and their removal is an intractable problem to be solved.

In recent years, a variety of methods have been developed to remove heavy metals from the environment: chemical precipitation, ion-exchange, adsorption and rhizofiltration. Chemical precipitation is a simple and low-cost process that is used extensively. However, it is not feasible when the metal concentration is low and it can generate great quantities of problematic sludge (Fu & Wang 2011). Ion-exchange can effectively remove high concentrations of heavy metals, but is costly, can cause secondary pollution and is also not feasible when the metal concentration is low (Athanasiadis & Helmreich 2005). Adsorption is an effective method to remove heavy metals at low concentrations, although its efficiency depends on the type and suitability of the adsorbent (Babel & Kurniawan 2003). Rhizofiltration uses plants to remove heavy metals from the environment; however, the efficiency is typically low (Dushenkov *et al*. 1995). Recently, microalgae have been studied to remove heavy metals from the environment (Lukač & Aegerter 1993; Zhou *et al*. 2006); however, it is difficult to harvest the metal-containing microalgae. All of these proposed methods have various disadvantages and it is imperative to find an effective and low-cost method to remove heavy metals from the environment.

Biosorption of heavy metals is a relatively new technology for their remediation. Macroalgae are widely distributed, fast-growing and grow all year round and are known to absorb heavy metals from the environment. *U. prolifera* is a globally common species of macroalgae with tremendous biomass. It is well-known for causing the green tides in Qingdao, China since 2007, and for affecting the Olympic Games sailing competition in 2008. It was estimated that more than 1.5 million wet tonnes of thalli were salvaged and around 20

million wet tonnes were produced along the shores of Qingdao in 2008 (Gao *et al*. 2011). Additionally, some species of *Ulva* can grow in freshwater. These characteristics of *U. prolifera* indicate its potential as a novel material for removing heavy metals from the aquatic environment.

In this study, we aimed to determine the ability of fresh *U. prolifera* thalli to remove additional Cd^{2+} in several kinds of water backgrounds: seawater, freshwater, water containing domestic garbage, water containing industrial waste and the elutriant of soil. The Cd^{2+} absorption capacity was evaluated and the effect of Cd^{2+} on photosynthetic parameters in the thalli was assessed.

Materials and methods

Sample collection

Thalli of *U. prolifera* were collected from the intertidal zone at Huiquan Bay (36°0′3″N, 121°2′0″E), Qingdao, China. The thalli were rinsed in sterile seawater to remove sediment and epiphytes, and then cultivated at 20°C with 100 μmol photons m^{-2} s^{-1} irradiance for further study.

Cd^{2+} treatment in seawater

Thalli (4 g fresh weight, FW) were cultured in 250-mL flasks containing 200 mL of culture solution (seawater). The Cd^{2+} concentrations were set as 0 (control), 1, 10 and 100 μM with the addition of Cd^{2+} stock solution, prepared by dissolving analytical grade $CdCl_2$ (Tianjin Guangfu Fine Chemical Research Institute, Tianjin, China) in pure water. Each concentration was replicated four times. The experiment was conducted at 20°C with 100 μmol m^{-2} s^{-1} irradiance for 12 h each day. The flasks were shaken several times daily. The photosynthetic activity of thalli cultivated in different Cd^{2+} concentrations was measured at 1, 2, 3, 4 and 5 days after treatment initiation. After photosynthetic activity was measured, 2 mL of culture solution was taken to measure the remaining Cd^{2+} concentration. In addition, after treatment for 5 days, thalli were transferred to seawater without Cd^{2+} to culture for another 3 days and their photosynthetic activities were again measured.

Cd^{2+} treatment in freshwater

The set of Cd^{2+} concentrations and the culture condition of thalli treated in freshwater were the same as for Cd^{2+} treatment in seawater. Photosynthetic activity was measured at 1, 2 and 3 days after treatment initiation. Two millilitre culture solution was taken to measure the remaining Cd^{2+} concentration after each photosynthetic activity measurement. In addition, after treatment for 3 days, thalli were transferred to freshwater without Cd^{2+} to culture for another 3 days and their photosynthetic activities were also measured.

Cd^{2+} treatment in water containing domestic garbage and industrial waste

To check the Cd^{2+} absorption capacity of *U. prolifera* in natural environment, some water was collected from Haibo River (36°5′50N, 120°21′15″E), Qingdao, China. The river flows through the city and contains domestic garbage. In addition, some water from Liaocheng city (36°2′2″N, 115°65′0″E), Shandong Province, was also collected. Some factories were located in the collection site. Accordingly, the water likely contained some industrial waste. The contents of main ions in the water samples were determined (Supporting Information Table 1). The Cd^{2+} concentration of water containing domestic garbage was only 0.005 μg L^{-1} while 0.13 μg L^{-1} in water containing industrial waste, both lower than the environmental quality standards for surface water (10 μg L^{-1}, GB 3838–2002). Additional Cd^{2+} was added to the water. The set of Cd^{2+} concentrations and culture environment were the same as for Cd^{2+} treatment in freshwater. For the thalli cultured in water containing domestic garbage, photosynthetic activity was measured at 1, 2 and 3 days after treatment initiation. For the thalli cultured in water containing industrial waste, photosynthetic activity was measured at 1 and 2 days after treatment initiation. After photosynthetic activity was measured, 2 mL of culture solution was taken to measure the remaining Cd^{2+} concentration.

Cd^{2+} treatment in the elutriant of soil

In addition to contaminating water, metals can also contaminate soil. To detect the Cd^{2+} absorption capacity of *U. prolifera* in soil, some soil was collected from Jimo city (36°21′12″N, 120°39′54″E), Shandong Province, to make artificially polluted soil. The final Cd^{2+} concentrations in the soil were set as 0, 50, 100, 200 and 300 mg kg^{-1}. Elutriant of soil was made using 500 mL of 0.2 M oxalic acid, 500 mL of 0.02 M ethylenediaminetetraacetic acid and 50 g of soil. NaOH was used to adjust pH to 7. The respective Cd^{2+} concentrations of the elutriant were measured at about 0, 10, 15, 40 and 71 mg L^{-1}. Thalli (4 g FW) of *U. prolifera* were put in flasks containing 200 mL of elutriant in four replicates. The culture environment was the same as for Cd^{2+} treatment in freshwater. Photosynthetic activity was measured after 1 day treatment and the remaining Cd^{2+} concentration was also tested.

Photosynthetic activity and Cd^{2+} absorption capacity measurement

The photosynthetic activity of the thalli was measured at 20°C using a Dual pulse-amplitude-modulated (PAM) fluorometer (Heinz Walz, Effeltrich, Germany) connected to a computer with WinControl software. The measurement method was followed as described by Gao *et al*. (2011).

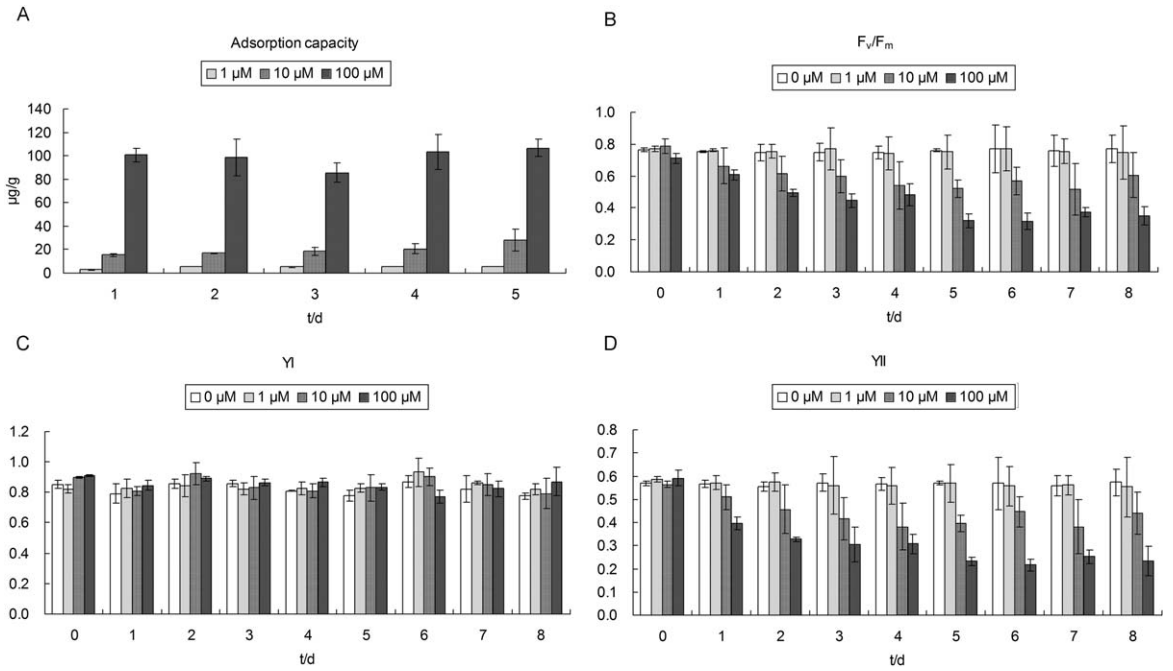


Fig. 1. Cd^{2+} absorption capacity and photosynthetic parameters (F_v/F_m, YI and YII) change of *U. prolifera* cultivated in seawater with different Cd^{2+} concentrations. The first 5 days are treatment time while the last 3 days are recovery time. (a) Cd^{2+} absorption capacity; (b) F_v/F_m; (c) YI; (d) YII.

To detect the remaining Cd^{2+} concentration in the culture solution, 2 mL of solution was collected after the measurement of photosynthetic activity. The analysis was conducted using an Optima 7300DV inductively coupled plasma optical emission spectrometry (Perkin-Elmer, USA). The Cd^{2+} absorption capacity of the thalli was calculated according to the formula: $(m_0 - m_e)/m_t$, where m_0 is the initial Cd^{2+} mass in the solution (μg) and m_e is the remaining Cd^{2+} mass (μg), while m_t is the mass of the *U. prolifera* thalli (g) in the system.

Results

Cd^{2+} absorption capacity and photosynthetic parameters change for *U. prolifera* cultivated in seawater with different Cd^{2+} concentrations

When the Cd^{2+} concentration in seawater was low (1 μM), the absorption capacity of the thalli increased significantly from 1 day to 2 days, and with no significant changes in the following days. In the 10 and 100 μM treatments, it did not change significantly with prolonged treatment time. However, the Cd^{2+} absorption capacity increased significantly as the increase of Cd^{2+} concentration (Fig. 1a, Table 1).

Figure 1(b)–(d) shows the values of photosynthetic parameters of *U. prolifera* cultivated in seawater with different Cd^{2+} concentrations, where 1–5 days was Cd^{2+} treatment time and 6–8 days was recovery time (i.e. culture solution without Cd^{2+}). The values of F_v/F_m did not change significantly in the low Cd^{2+} concentration (1 μM) during the treatment and recovery periods. However, in the 10 and 100 μM treatments, F_v/F_m decreased significantly with prolonged treatment time and did not recover to normal values during recovery (Fig. 1b). A similar trend was found in YII (Fig. 1d). Interestingly, the value of YI did not significantly change during the treatment and recovery period in all Cd^{2+} treatments (Fig. 1c).

Table 1 Cd^{2+} absorption capacity of *U. prolifera* thalli treated in different aquatic environments 1 day after initiation

	Cd^{2+} Absorption (μg/g)		
Aquatic environment	1 μM	10 μM	100 μM
Seawater	2.787 ± 0.076	15.350 ± 1.213	100.633 ± 15.711
Freshwater	5.333 ± 0.111	45.825 ± 2.158	420.775 ± 12.619
Water containing domestic garbage	4.120 ± 0.173	45.517 ± 2.176	469.200 ± 32.138
Water containing industrial waste	3.720 ± 1.370	40.838 ± 3.363	385.550 ± 29.410

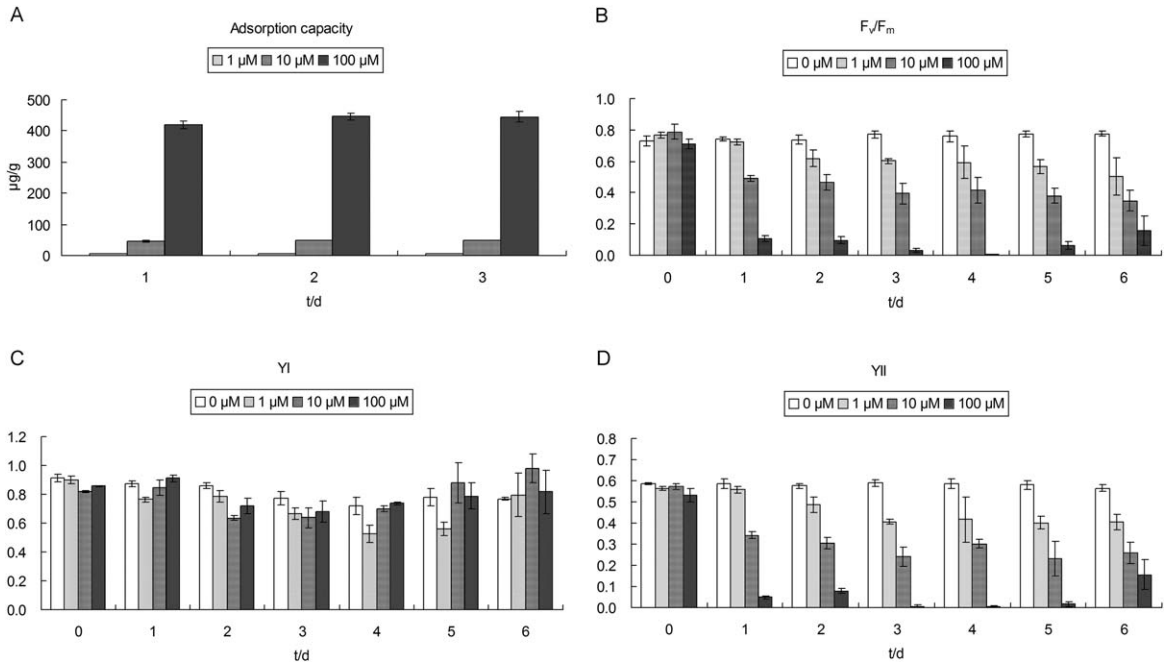


Fig. 2. Cd^{2+} absorption capacity and photosynthetic parameters (F_v/F_m, YI and YII) change of *U. prolifera* cultivated in freshwater with different Cd^{2+} concentrations. (a) Cd^{2+} absorption capacity; (b) F_v/F_m; (c) YI; (d) YII.

Cd^{2+} absorption capacity and photosynthetic parameters change for *U. prolifera* cultivated in freshwater with different Cd^{2+} concentrations

In each Cd^{2+} treatment, Cd^{2+} absorption capacity of the thalli did not change significantly with prolonged treatment time. However, it increased significantly as the increase of Cd^{2+} concentration (Fig. 2a). In addition, compared with the seawater treatments, the absorption capacity seemed stronger in the freshwater treatments (Table 1).

Figure 2(b)–(d) shows photosynthetic parameters changes of *U. prolifera* cultivated in freshwater with different Cd^{2+} concentrations, where 1–3 days was Cd^{2+} treatment time and 4–6 days was recovery time (i.e. culture solution without Cd^{2+}). F_v/F_m changed significantly during treatment period, especially in the 10 and 100 μM treatments. Although the thalli recovered for 3 days, F_v/F_m did not increase significantly although the 100 μM treatment showed a slight increase (Fig. 2b). A similar trend was found for YII (Fig. 2d). During the Cd^{2+} treatment period, YI had a minor decreasing trend and increased slightly when thalli were recovered (Fig. 2c).

Cd^{2+} absorption capacity and photosynthetic parameters change for *U. prolifera* cultivated in water containing domestic garbage or industrial waste with Cd^{2+} adding

Similar to freshwater treatments, Cd^{2+} absorption capacity of the thalli in water containing domestic garbage or industrial waste also did not change significantly with prolonged treatment time, and it increased significantly as the increase of Cd^{2+} concentration (Figs 3a and 4a). F_v/F_m decreased significantly during the treatment period, with more marked decrease for Cd^{2+} treatment groups than the control (Figs 3b and 4b). There was a similar trend for YII (Figs 3d and 4d). In water containing domestic garbage, YI had a decreasing trend in the experimental groups with prolonged treatment time; however, the decrease was not as great as for F_v/F_m and YII (Fig. 3c). In water containing industrial waste, YI did not change significantly during the treatment period in all Cd^{2+} treatments (Fig. 4c).These results suggested that some substance in the water had a negative influence on *U. prolifera* thalli, and Cd^{2+} still influenced the photosynthesis.

Cd^{2+} absorption capacity and photosynthetic parameters change for *U. prolifera* cultivated in elutriant of Cd^{2+}-added soil

When thalli were cultivated in the elutriant of Cd^{2+}-added soil, Cd^{2+} absorption capacity of the thalli increased significantly as the increase of Cd^{2+} concentration (Fig. 5a). F_v/F_m and YII decreased almost to zero in all treatments, including controls (Fig. 5b and d). However, YI showed a stronger tolerance than YII, although it decreased with increased Cd^{2+} content (Fig. 5c).

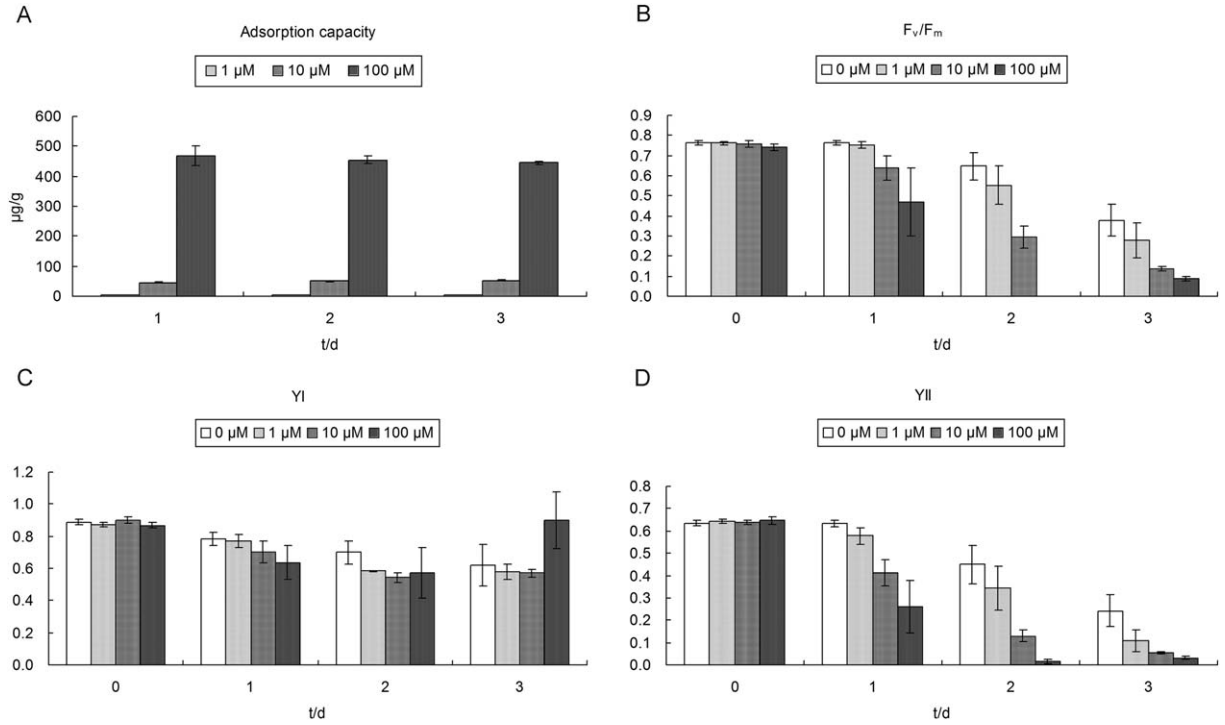


Fig. 3. Cd^{2+} absorption capacity and photosynthetic parameters (F_v/F_m, YI and YII) change of *U. prolifera* cultivated in water containing domestic garbage with different Cd^{2+} concentrations. (a) Cd^{2+} absorption capacity; (b) F_v/F_m; (c) YI; (d) YII.

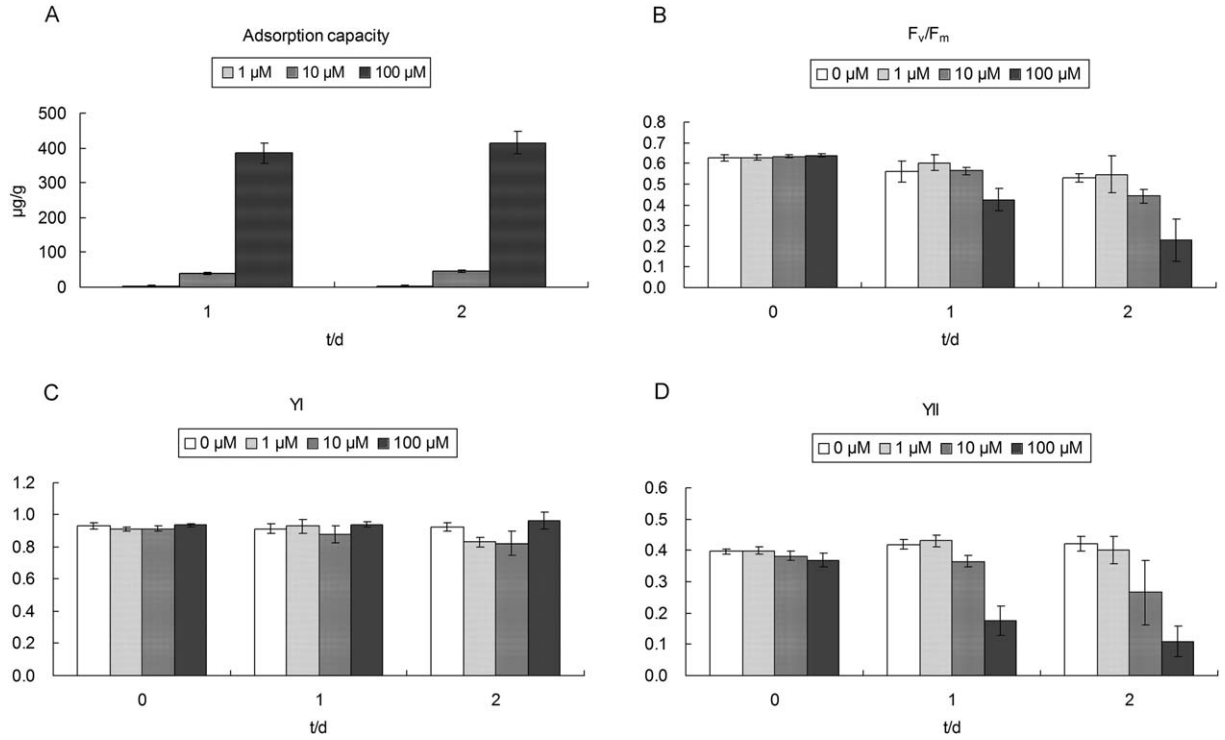


Fig. 4. Cd^{2+} absorption capacity and photosynthetic parameters (F_v/F_m, YI and YII) change of *U. prolifera* cultivated in water containing industrial waste with additional Cd^{2+}. (a) Cd^{2+} absorption capacity; (b) F_v/F_m; (c) YI; (d) YII.

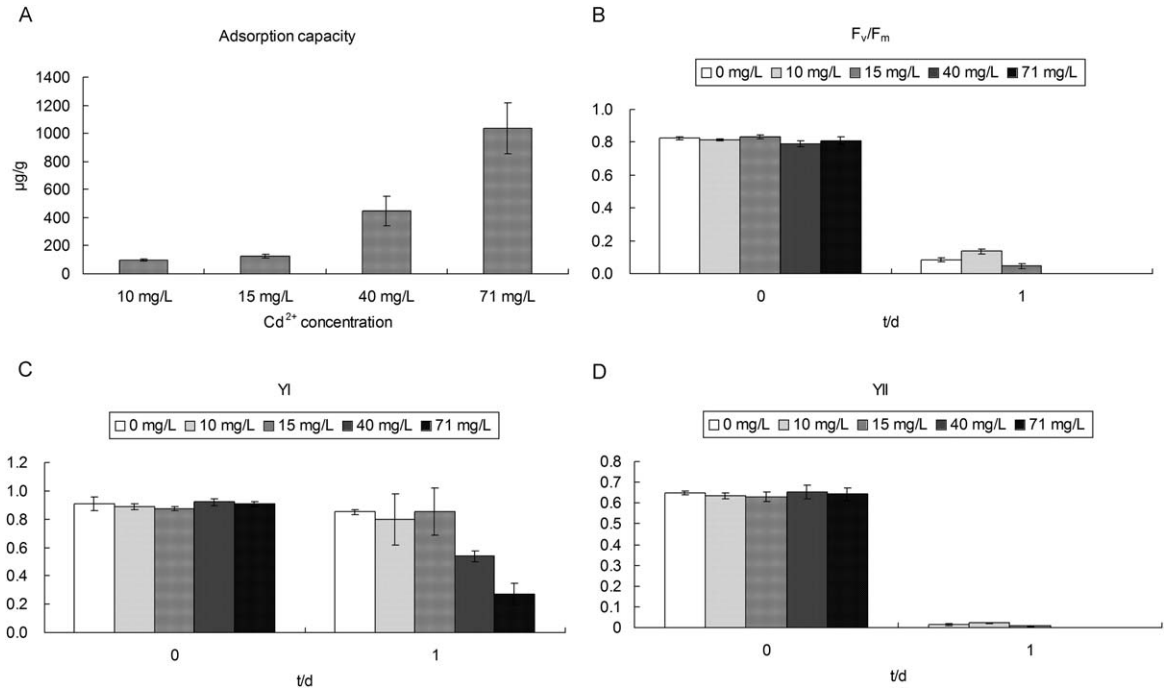


Fig. 5. Cd^{2+} absorption capacity and photosynthetic parameters (F_v/F_m, YI and YII) change of *U. prolifera* cultivated in elutriant of Cd^{2+} added soil. (a) Cd^{2+} absorption capacity; (b) F_v/F_m; (c) YI; (d) YII.

Discussion

The decrease of photosynthetic activity suggested that Cd^{2+} entered *U. prolifera* cells

When *U. prolifera* thalli were cultured in water environments with additional Cd^{2+}, photosynthetic activity decreased significantly, especially the values of F_v/F_m and YII. Compared with seawater at identical Cd^{2+} concentration, the Cd^{2+} absorption capacity was higher and photosynthetic activity decreased more significantly in freshwater, water containing domestic garbage or domestic garbage (Table 1), suggesting the change in photosynthetic activity was related to absorption capacity. Furthermore, with increased amount of Cd^{2+} absorbed by the thalli, the photosynthetic activity decreased, showing that Cd^{2+} was a considerable stress to the alga and had entered the cells. Thalli of *U. prolifera* comprise a single layer of cells, which is convenient for entry of Cd^{2+}.

Cd^{2+} appears to be absorbed passively and translocated freely (Cutler & Rains 1974; Jarvis *et al*. 1976). In particular, Cd^{2+} uptake seems to be in competition for the same transmembrane carrier with nutrients such as calcium (Ca), magnesium (Mg), iron (Fe) and manganese (Clarkson & Lüttge 1989; Rivetta *et al*. 1997; Benavides *et al*. 2005). It was reported that Cd was taken up by Ca and Fe transporters in *Micrasterias* (Volland *et al*. 2014). In higher plants, Cd^{2+} is taken up through roots. Most accumulated Cd^{2+} was strongly retained by roots in soybean (Cataldo *et al*. 1983). However, algae have no obvious organ differentiation and their cells can absorb Cd^{2+} from the aquatic environment. Cd^{2+} interaction is affected by pH and Ca^{2+} concentrations. The availability of Cd^{2+} appeared to be reduced by competition with low pH or high levels of Ca^{2+} (Christensen 1984; Prasad 1995). Compared with the other three water environments (freshwater, water containing domestic garbage and domestic garbage), seawater contains more divalent ions, such as Ca^{2+} and Mg^{2+}. We presume that in seawater the ion-binding sites are predominantly occupied, thus Cd^{2+} cannot bind easily to these sites and Cd^{2+} absorption capacity is relatively low. In addition, it was reported that Cd accumulation generally increased with falling salinity in *Carcinus maenas*, perhaps due to the reduction in apparent water permeability (Wright 1977; Burke *et al*. 2003), which might use for reference to explain the lower Cd^{2+} absorption capacity in seawater.

When *U. prolifera* thalli were treated with Cd^{2+}, PSI showed a stronger tolerance to the stress than PSII, represented by a greater decrease in YII than YI. This is consistent with the responses of macroalgae to other stresses, such as dehydration and high salinity (Gao *et al*. 2011; Huan *et al*. 2014). Cd^{2+} directly or indirectly inhibits many physiological processes, such as photosynthesis, respiration and gas exchange, probably due to the interaction with functional proteins and the inhibition or inactivation of many enzymes (Prasad *et al*. 2001; Faller *et al*. 2005). It was reported that Cd^{2+} may be preferentially accumulate in chloroplasts

(Lagriffoul *et al*. 1998) and PSI activity was slightly affected by Cd^{2+} (Atal *et al*. 1991; Siedlecka & Krupa 1996). However, in many cases, Cd^{2+} inhibits PSII, possibly affecting the water splitting system and ultimately causing the disassembly of PSII (Atal *et al*. 1991). In the cyanobacterium *Microcystis aeruginosa*, the inhibitory site of Cd was probably situated on the ferredoxin/$NADP^+$-oxidoreductase enzyme at the terminal of the whole electron transport chain (Zhou *et al*. 2006). In the present study, F_v/F_m and YII decreased significantly while PSI did not change significantly with increased Cd^{2+} concentration, thus we presume that the inhibitory site of Cd^{2+} in *U. prolifera* was located at PSII level. In addition, the damage caused by Cd^{2+} was irreparable in the tested period, for the photosynthetic activity of Cd-treated thalli did not recover during the recovery period (Figs 1 and 2).

Cd^{2+} absorption of *U. prolifera* has a eurytopic environmental background

Cd^{2+} is a nonessential, nonbeneficial element with a high toxic potential. Anthropogenic activities, such as metal industry, release of waste and sewage water, mining, production, use and disposal of batteries and the use of fertilizers and pesticides, can lead to the dispersion of Cd (Hanikenne 2003). Furthermore, compared to other metals, Cd has high solubility in water, which determines its wide distribution in aquatic ecosystems (Lockwood 1976). In the present study, several kinds of water environments were used as backgrounds: seawater, freshwater, water containing domestic garbage, water containing industrial waste and the elutriant of soil. With additional Cd^{2+} in the water backgrounds, the Cd^{2+} absorption capacity of *U. prolifera* was relatively high, although with some differences. The results suggested that Cd^{2+} absorption of *U. prolifera* had a eurytopic environmental background.

Ulva is an economically important macroalga, extremely common in marine environments and with wide tolerance to salt, temperature and dehydration – showing strong adaptability to the environment (Gao *et al*. 2011; Huan *et al*. 2014). It was reported that *Enteromorpha* could survive in freshwater for 3–5 weeks, and its growth increased with higher salinity within a certain range (Edwards *et al*. 1988; Kamer & Fong 2000). In addition, free-floating *U. prolifera* is one of the causative species of green tides. When green tides occur, massive mats of floating thalli accumulate rapidly in surface seawater with huge biomass. All of these reasons make *U. prolifera* a suitable material to remove Cd^{2+} from aquatic environments.

Conclusions

(1) Our results indicate that *U. prolifera* is a good material to remove Cd^{2+} from the aquatic environment, including seawater, freshwater, water containing domestic garbage, water containing industrial waste and elutriant of soil with additional Cd^{2+}.

(2) In most cases, the Cd^{2+} absorption capacity of the thalli was relatively high.

(3) In addition, the propensity for *U. prolifera* to have a large biomass can provide the quantities required for remediating large polluted water bodies.

(4) The use of *U. prolifera* to remove Cd^{2+} in the environment is competitive with existing technologies such as chemical precipitation, ion-exchange adsorption, rhizofiltration and membrane filtration.

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